

Learning4help by Waqas Qureshi

Asslam-o-Alaikum! In this file we'll provide you **CS301 Important Midterm MCQs. Must Prepare before Midterm Exams.**
Verify Answers yourself too. I am not responsible for any answer. If you found mistake then inform me.
If you found mistake then let us know.

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Question # 1 of 10 (Start time: 11:10:29 PM, 12 November 2024) Total Marks: 1

Stack can be implemented by which of the following data structure?

Select the correct option

- ☒ Array
- ☐ Skip list
- ☐ BST
- ☐ AVL

There is no such node whose next field is NULL, which one of the given options supports the statement.

Select the correct option

- ☐ Linked list
- ☐ Queue
- ☒ Circular link list
- ☐ Array

Which of the following is the correct conversion of infix to postfix expression?
 $Z+B-(D-H)/K$

Select the correct option

- ☐ $ZB+DH-K- /$
- ☒ $ZB+DH-K / -$
- ☐ $ZB+D-H-K /$
- ☐ $ZB+DHK-- /$

Which of the following is the correct conversion of infix to postfix expression?
 $Z+B-(D-H)/K$

Select the correct option

- ☐ $ZB+DH-K- /$
- ☒ $ZB+DH-K / -$
- ☐ $ZB+D-H-K /$
- ☐ $ZB+DHK-- /$

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Question # 4 of 10 (Start time: 11:12:35 PM, 12 November 2024) Total Marks: 1

If numbers 5, 222, 4, 48 are inserted in a queue, which one will be removed first?

Select the correct option

- ☐ 1
- ☒ 5
- ☐ 222
- ☐ 48

Question # 5 of 10 (Start time: 11:12:52 PM, 12 November 2024) Total Marks: 1

Local variables of a function are stored in,

Select the correct option

- ☒ Stack
- ☐ Queue
- ☐ Binary Search Tree
- ☐ AVL Tree

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Question # 6 of 10 (Start time: 11:13:06 PM, 12 November 2024) Total Marks: |

_____ is the major factor to see the efficiency of a program.

Select the correct option

- ☒ Time
- ☐ Quality
- ☐ Correctness
- ☐ Code

Question # 7 of 10 (Start time: 11:13:32 PM, 12 November 2024) Total Marks: |

If we use array to implement list, then there is an issue that it gives difficulty when

Select the correct option

- ☐ We will remove data from it
- ☐ We will decrease its size
- ☐ We will access values randomly
- ☒ We will increase its size

Question # 8 of 10 (Start time: 11:13:59 PM, 12 November 2024) Total Marks: 1

Generalized code written for a class is called:

Select the correct option

- ☐ Structure
- ☐ Function
- ☐ Stack
- ☒ Template

Question # 9 of 10 (Start time: 11:14:27 PM, 12 November 2024) Total Marks: 1

Which operation of queue data structure is used to insert an element into the Queue?

Select the correct option

- ☐ remove()
- ☐ front()
- ☒ enqueue()
- ☐ dequeue()

Which of the following is the limitation of the array?

Select the correct option

- ☐ Cannot store more than 1000 elements
- ☒ Fix size
- ☐ Have no random access
- ☐ Cannot store user define objects

Question # 1 of 10 (Start time: 11:19:36 PM, 12 November 2024) Total Marks: 1

In _____ various cells of memory are not located continuously.

Select the correct option

- ☐ Arrays
- ☒ Linked list
- ☐ list
- ☐ Circular array

Int * i = new int {10};
Above given code will:

Select the correct option

- ☒ Allocate memory for 10 integers
- ☐ Create 10 pointers of integer type
- ☐ Create an integer having value 10
- ☐ Allocate memory for 9 integers

copy() method of List data structure -----

Select the correct option

- ☐ Copy last item of list
- ☐ Copy any item of list
- ☒ Set one list to be a copy of another
- ☐ Copy first item of list

Question # 4 of 10 (Start time: 11:20:48 PM, 12 November 2024) Total Marks: 1

Each operator in a postfix expression refers to the previous _____ operand(s).

Select the correct option

- ☐ one
- ☐ four
- ☒ two
- ☐ three

Consider the linked list having data [6, 72, 35, 65, 25] stored in it. While current pointer is pointing to memory location having 72 stored in it. After calling remove() function on the following linked list current point will point to memory location having value?

Select the correct option

- ☒ 35
- ☐ 25
- ☐ 6
- ☐ 65

Question # 6 of 10 (Start time: 11:21:30 PM, 12 November 2024) Total Marks: 1

Consider the linked list having data [6, 72, 35, 65, 25] stored in it. While current pointer is pointing to memory location having 72 stored in it. After calling add(4) function on the following linked list current point will point to memory location having value?

Select the correct option

- ☐ 4
- ☐ 35
- ☐ 25
- ☐ 72

Consider the following infix expression.
 $7/8 + 9$
If one converts the above expression into postfix, what would be the resultant expression?

Select the correct option

- ☐ 7 8 9 / +
- ☐ / 7 8 + 9
- ☒ 7 8 / + 9
- ☐ 7 8 / 9 +

Which of the following statement is correct for the variable "current--"?

Select the correct option

- ☐ current = current + 1
- ☐ current = current + 2
- ☒ current = current - 1
- ☐ current - 1 = current

Suppose you are writing a class for Node class and forgot to write the constructor of the class, then

Select the correct option

- ☐ Compiler will give run time error
- ☐ Compiler will do nothing
- ☒ Compiler will automatically generate a default constructor
- ☐ Compiler will give compile time error

Question # 10 of 10 (Start time: 11:23:24 PM, 12 November 2024) Total Marks: |

To search an element which method of the list searches the entire list from beginning to end?

Select the correct option

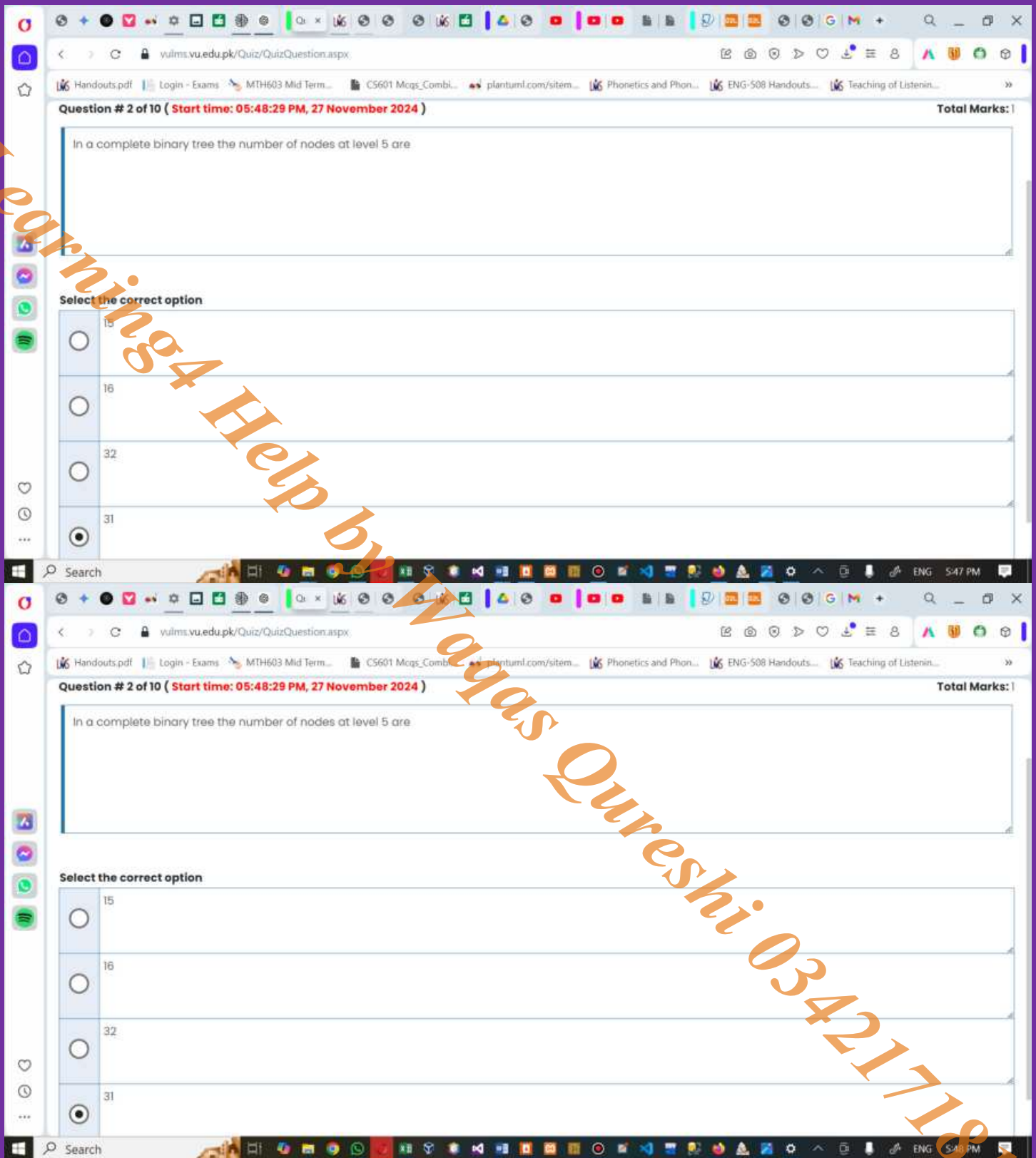
- ☐ update()
- ☐ end();
- ☐ start();
- ☒ find();

Question # 1 of 10 (Start time: 05:48:04 PM, 27 November 2024) Total Marks: |

Local variables defined inside function body are _____ automatically at the end of function execution.

Select the correct option

- ☐ created
- ☐ incremented
- ☒ destroyed
- ☐ decremented



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Question # 3 of 10 (Start time: 05:49:32 PM, 27 November 2024) Total Marks: 1

Following is a keyword of C++ _____

Select the correct option

- ☐ add
- ☐ eliminate
- ☐ remove
- ☒ delete

Question # 4 of 10 (Start time: 05:49:54 PM, 27 November 2024) Total Marks: 1

Function signatures are also called :

Select the correct option

- ☐ Function definition
- ☐ Function overloading
- ☐ Function overriding
- ☒ Function prototype

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Question # 5 of 10 (Start time: 05:50:40 PM, 27 November 2024) Total Marks: 1

If we return the reference of a local variable from a function it will cause :

Select the correct option

- ☐ deletion of local variable from memory
- ☐ reference overloading
- ☐ duplication of local variable
- ☒ dangling reference

Search

Question # 6 of 10 (Start time: 05:51:02 PM, 27 November 2024) Total Marks: 1

A _____ is a tree in which every level, except possibly the last, is completely filled.

Select the correct option

- ☐ Strict Binary tree
- ☐ Perfect Binary tree
- ☐ Full Binary Tee
- ☒ Complete binary tree

Search

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CS301 - Data Structures (Quiz No.2) Quiz Start Time: 05:48 PM, 27 November 2024

Question # 7 of 10 (Start time: 05:51:39 PM, 27 November 2024) Total Marks: 1

A node in a binary tree can have maximum _____ child node(s).

Select the correct option

Reload Math Equations

☐ 1

☐ 0

☐ 3

☒ 2

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Question # 8 of 10 (Start time: 05:51:54 PM, 27 November 2024) Total Marks: 1

In tree, the search operation is _____ as compared to the linked list.

Select the correct option

☒ very fast

☐ equally time-taken

☐ not fast

☐ very slow

Search

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CS301 - Data Structures (Quiz No.2) Quiz Start Time: 05:48 PM, 27 November 2024

Question # 9 of 10 (Start time: 05:52:17 PM, 27 November 2024) Total Marks: 1

In Binary Search Tree, deleting a node is easy if it is a _____ node.

Select the correct option

☒ Leaf

☐ Root

☐ Right

☐ Left

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CS301 - Data Structures (Quiz No.2) Quiz Start Time: 05:48 PM, 27 November 2024

Question # 10 of 10 (Start time: 05:52:42 PM, 27 November 2024) Total Marks: 1

Which of the following is TRUE about recursive calls ?

Select the correct option

☐ Recursive calls can only be used with inorder traversal

☒ Recursive call always contains a terminating condition

☐ There is no terminating condition in recursive calls

☐ Tree data structure cannot use recursive calls

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Question # 1 of 10 (Start time: 07:54:47 PM, 27 November 2024) Total Marks: |

Whenever a node is deleted from a binary search tree, _____ has to be maintained.

Select the correct option

- ☒ In order traversal
- ☐ Any traversal can be maintained
- ☐ Pre order traversal
- ☐ Post order traversal

CS301 - Data Structures (Quiz No.2) Quiz Start Time: 07:54 PM, 27 November 2024

Question # 2 of 10 (Start time: 07:55:21 PM, 27 November 2024) Total Marks: |

Two common models of simulation are _____ and _____

Select the correct option

- ☐ Circuit-based simulation and Time-based simulation
- ☐ Circuit-based simulation and Event-based simulation
- ☒ Time-based simulation and Event-based simulation
- ☐ Circuit based simulation and Object oriented simulation

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CS301 - Data Structures (Quiz No.2) Quiz Start Time: 07:54 PM, 27 November 2024

Question # 3 of 10 (Start time: 07:55:58 PM, 27 November 2024) Total Marks: 1

To create a reference variable, we need to use _____ sign.

Select the correct option

☐ dollar

☐ sigma

☒ ampersand

☐ asterisk

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Analyze the given code carefully and identify which type of binary tree traversal this is :

```
void traversal (TreeNode<int>* treeNode)
{
    if( treeNode != NULL )
    {
        cout << *(treeNode->getInfo())<<" ";
        traversal (treeNode->getLeft());
        traversal (treeNode->getRight());
    }
}
```

Select the correct option

☒ Preorder

☐ Inorder

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Question # 5 of 10 (Start time: 07:56:49 PM, 27 November 2024) Total Marks: 1

-BinarySearchTree() is a _____

Select the correct option

- ☐ Switch Case
- ☐ Template method call
- ☒ Destructor
- ☐ Constructor

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Question # 6 of 10 (Start time: 07:57:11 PM, 27 November 2024) Total Marks: 1

From Operating System point of view, the recursive function calls are made with the help of _____

Select the correct option

- ☐ Linked list
- ☒ Stack
- ☐ Binary Search Tree
- ☐ Queue

Search

What will be level order traversal of the given BST ?

```
graph TD; 6((6)) --- 2((2)); 6 --- 8((8)); 2 --- 1((1)); 2 --- 5((5)); 5 --- 3((3)); 3 --- 4((4));
```

Select the correct option

☐ 1 2 3 4 5 6 8

☒ 6 2 8 1 5 3 4

☐ 1 4 3 5 2 8 6

☐ 1 2 3 4 5 6 8

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Which of the following function(s) is/are used for function call by value

```
void Func1(int& num) {  
    num++;  
}  
  
void Func2(int num) {  
    num++;  
}
```

Select the correct option

☐ C++ do not have concept of function call by value

☐ Func1

☐ Both Func1 and Func2

☒ Func2

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Question # 9 of 10 (Start time: 07:59:25 PM, 27 November 2024) Total Marks: 1

_____ is used for Reference variables in C++.

Select the correct option

- ☒ &
- ☐ !
- ☐ @
- ☐ #

Question # 10 of 10 (Start time: 07:59:48 PM, 27 November 2024) Total Marks: 1

Which of the following is TRUE for search operations in a binary tree as compared to linked list or an array?

Select the correct option

- ☐ It increases the number of comparisons
- ☒ It reduces the number of comparisons
- ☐ The number of comparisons remain same
- ☐ Search operation becomes slower

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CS301 - Data Structures (Quiz No.2) Quiz Start Time: 10:14 PM, 29 November 2024

Question # 2 of 10 (Start time: 10:15:06 PM, 29 November 2024) Total Marks: 1

_____ objects (objects accessed by pointers) are called anonymous objects.

Select the correct option

☐ Public

☐ Friend

☐ Private

☒ Nameless

Reload Math Equations

Question # 3 of 10 (Start time: 10:15:50 PM, 29 November 2024) Total Marks: 1

Suppose the elements are "3","5","1","6", If we want to enqueue "8" then dequeue all the elements, what will be the order of elements?

Select the correct option

☒ 61538

☐ 35168

☐ 86153

☐ 83516

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Question # 4 of 10 (Start time: 10:17:03 PM, 29 November 2024) TOTAL MARKS: 1

What will be the preorder traversal of the given tree?

```
graph TD; 45((45)) --> 25((25)); 45 --> 75((75)); 25 --> 15((15)); 25 --> 35((35));
```

Select the correct option

☐ 45 75 15 35 25

☐ 45 25 75 15 35

☒ 45 25 15 35 75

☐ 15 35 25 45 75

Reload Math Equations

CS301 - Data Structures (Quiz No.2) Quiz Start Time: 10:14 PM, 29 November 2024

Question # 6 of 10 (Start time: 10:19:12 PM, 29 November 2024) Total Marks: 1

A complete binary tree has 63 non-leaf nodes. The number of leaf nodes will be:

Select the correct option

☐ 62

☐ 61

☒ 64

☐ 65

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CS301 - Data Structures (Quiz No.2) Quiz Start Time: 10:14 PM, 29 November 2024

Question # 8 of 10 (Start time: 10:21:00 PM, 29 November 2024) Total Marks: 1

In which traversal method, the recursive calls can be used to traverse a binary tree ?

Select the correct option

☐ In postorder and inorder traversal only

☐ In preorder and inorder traversal only

☐ In postorder and preorder traversal only

☒ In preorder, postorder and inorder traversal only

Reload Math Equations

Left, right, info, and parent are the operations of _____ data structure.

Select the correct option

☐ Linked List

☐ Stack

☒ Tree

☐ Queue

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Question # 10 of 10 (Start time: 10:21:50 PM, 29 November 2024) Total Marks: 1

A complete binary tree has 16 leaf nodes. The number of non-leaf nodes will be:

Select the correct option

☒ 15

☐ 13

☐ 16

☐ 14

Click to Save Answer & Move to Next Question

Question # 1 of 10 (Start time: 06:38:50 AM, 07 May 2024) Total Marks: 1

When an operator is found in stack while evaluating a postfix expression, which of the following operation is performed on previous two operands?

Select the correct option

☐ Enqueue

☐ None of the given options

☐ push

☒ pop

Question # 2 of 10 (Start time: 06:40:04 AM, 07 May 2024) Total Marks: 1

When the call to remove a node is made, a node which is being pointed by _____ will be deleted.

Select the correct option

Reload Math Equations

☐ Head Node

☐ Null node

☐ Last Current node

☒ Current node

Click to Save Answer & Move to Next Question

Question # 3 of 10 (Start time: 06:41:54 AM, 07 May 2024) Total Marks: 1

Which type of linked list has two Null pointers?

Select the correct option

Reload Math Equations

☐ Circularly linked list

☒ Doubly linked list

☐ Singly linked list

☐ None of the given options

Click to Save Answer & Move to Next Question

Question # 4 of 10 (Start time: 06:42:20 AM, 07 May 2024) Total Marks: 1

Which of the following Linked List method moves the pointer forward?

Select the correct option

☐ start ()

☐ remove ()

☐ add ()

☒ next ()

Click to Save Answer & Move to Next Question

Question # 5 of 10 (Start time: 06:43:16 AM, 07 May 2024) Total Marks: 1

In C++, a single stack can be used for different types by using _____.

Select the correct option

☒ Templates

☐ Linked Lists

☐ Postfix expressions

☐ Static variables

Saving...

Question # 6 of 10 (Start time: 06:44:22 AM, 07 May 2024) Total Marks: 1

Which of the following statement declares the Stack object of float data type if a Stack class is defined using C++ template?

Select the correct option

☒ `<float> Stack stack;`

☐ `Stack stack;`

☐ `Stack float stack;`

☐ `Stack <float> stack;`

Click to Save Answer & Move to Next Question

CS301P - Data Structures (Practical) (Regular Quiz 1) Quiz Start Time: 06:38 AM, 07 May 2024

Question # 7 of 10 (Start time: 06:46:06 AM, 07 May 2024) Total Marks: 1

The result of the postfix expression 632*+ is _____

Select the correct option

☐ 12

☒ 20

☐ 18

☐ 30

Question # 8 of 10 (Start time: 08:47:16 AM, 07 May 2024) Total Marks: 1

What is the output of the following program?

```
int main ()  
{  
    int array [5] = {10,20,30,40,50};  
    cout << array [3];  
    return 0;  
}
```

Select the correct option

☐ 30

☒ 40

☐ Syntax Error

Reload Math Equations

Question # 9 of 10 (Start time: 08:47:55 AM, 07 May 2024) Total Marks: 1

In stack, which node will be considered as the Top of stack?

Select the correct option

☐ Any node

☐ Last node

☐ Middle node

☒ First node

Reload Math Equations

Click to Save Answer & Move to Next Question

CS301P - Data Structures (Practical) (Regular Quiz 1) Quiz Start Time: 06:38 AM, 07 May 2024

Question # 10 of 10 (Start time: 06:48:20 AM, 07 May 2024) Total Marks: 1

Which of the following statement declares an array 'x' with 6 integers?

Select the correct option

☐ float x[6];

☐ int [6] x;

☐ int x = 6;

☒ int x[6];

Trying to remove an element from an empty stack is called _____

Select the correct option

☐ Overflow of stack

☐ Garbage collection

☒ Underflow of stack

☐ Empty collection

Question # 2 of 10 (Start time: 06:50:33 AM, 07 May 2024) Total Marks: 1

In queue data structure elements are removed from _____

Select the correct option

☐ Push

☐ Pop

☒ Front

☐ Rear

Click to Save Answer & Move to Next Question

Question # 3 of 10 (Start time: 06:50:33 AM, 07 May 2024) Total Marks: 1

_____ is a collection of elements arranged in a linear order.

Select the correct option

☐ Binary Search Tree

☐ Heap

☒ Stack

☐ AVL Tree

Consider the following expression:

$P = A * B + C - D$

In the above expression "+" is a _____ operator,

Select the correct option

- ☐ Technical
- ☐ Unary
- ☒ Binary
- ☐ Logical

copy() method of List data structure _____

Select the correct option

- ☐ Copy first item of list
- ☐ Copy any item of list
- ☐ Copy last item of list
- ☒ Set one list to be a copy of another

Which of the following statement is correct for the variable "current--"?

Select the correct option

- ☒ current = current - 1
- ☐ current - 1 = current
- ☐ current = current + 1
- ☐ current = current - 2

Which of the following statement is correct for the variable "current--"?

Select the correct option

- ☒ current = current - 1
- ☐ current - 1 = current
- ☐ current = current + 1
- ☐ current = current - 2

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Question # 7 of 10 (Start time: 06:55:59 AM, 07 May 2024) Total Marks: 1

The _____ method of list will position the currentNode and lastCurrentNode at the start of the list.

Select the correct option

- ☐ First
- ☐ Remove
- ☐ Back
- ☒ Start

Question # 8 of 10 (Start time: 06:57:40 AM, 07 May 2024) Total Marks: 1

Which of the following is not a data structure ?

Select the correct option

- ☐ Stack
- ☒ Memory cell
- ☐ Queue
- ☐ Linked list

A software solution is said to be efficient if it solves the problem _____.

Select the correct option

- ☐ by consuming more hardware resources.
- ☐ by using some extra resources.
- ☒ within its resource constraints.
- ☐ within no time.

CS301 - Data Structures (Graded Quiz No. 1) Quiz Start Time: 06:49 AM, 07 May 2024

Question # 10 of 10 (Start time: 06:59:23 AM, 07 May 2024) Total Marks: 1

In the statement template <T>, T represents _____.

Select the correct option

- ☐ String data type
- ☐ Character data type
- ☒ Generic data type
- ☐ Integer data type

Reload Math Equations

Level-order traversal for Binary Search Tree can be implemented, _____.

Select the correct option

- ☐ Through max-heap
- ☐ Through both recursive method call and non-recursive method call
- ☒ Only through non-recursive method
- ☐ Only through recursive method

Factorial is an example of _____ function.

Select the correct option

- ☐ Cube
- ☒ Recursive
- ☐ Non-Recursive
- ☐ Log

CS301 - Data Structures (Graded Quiz No. 2) Quiz Start Time: 05:24 PM, 22 May 2024

Question # 3 of 10 (Start time: 05:26:00 PM, 22 May 2024) Total Marks: 1

Two common models of simulation are _____ and _____.

Select the correct option

Reload Math Equations

☐ Circuit-based simulation and Event-based simulation

☐ Circuit-based simulation and Time-based simulation

☒ Time-based simulation and Event-based simulation

☐ Circuit based simulation and Object-oriented simulation

CS301 - Data Structures (Graded Quiz No. 2) Quiz Start Time: 05:24 PM, 22 May 2024

Question # 4 of 10 (Start time: 05:26:28 PM, 22 May 2024) Total Marks: 1

In Binary Search Tree, deleting a node is easy if it is a _____ node.

Select the correct option

Reload Math Equations

☐ Right

☒ Leaf

☐ Left

☐ Root

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CS301 - Data Structures (Graded Quiz No. 2) Time Left: 05:24 PM, 22 May 2024

Question # 5 of 10 (Start time: 05:27:53 PM, 22 May 2024) Total Marks: 1

For searching a particular number in Binary Search Tree (if it is not present), the maximum number of comparisons will be _____ comparison(s) at each level.

Select the correct option Reload Math Equations

☐ 4

☐ 2

☒ 1

☐ 3

Elements in a queue data structure are added from _____ and removed from _____

Select the correct option

☐ Rear end

☒ Front end, rear end

☐ Rear end, front end

☐ Front end

When a function calling itself is called as _____

Select the correct option

- ☐ Nested loop
- ☐ Iteration
- ☐ Inline
- ☒ Recursion

Following is a keyword of C++ _____

Select the correct option

- ☒ delete
- ☐ eliminate
- ☐ remove
- ☐ del

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Question 1 of 10

In Preorder traversal of a binary tree the second step is _____

Select the correct option

- ☒ traverse the root
- ☐ traverse the left subtree
- ☐ traverse the right subtree
- ☐ traverse the leaf node

Question 1 of 10

In Preorder traversal of a binary tree the second step is _____

Select the correct option

- ☒ traverse the root
- ☐ traverse the left subtree
- ☐ traverse the right subtree
- ☐ traverse the leaf node

CS301 - Data Structures (Graded Quiz No. 2) Quiz Start Time: 12:05 AM, 23 May 2024

Question # 1 of 10 (Start time: 12:05:54 AM, 23 May 2024) Total Marks: 1

Which of the following calling method does not change the original value of variable passed to a function as an argument?

Select the correct option

Call by passing as pointer

Call by passing as value

Call by passing as reference

Call by passing as name

Reload Math Equations

Question # 2 of 10 (Start time: 12:06:27 AM, 23 May 2024) Total Marks: 1

After deletion of a node from a binary search tree, _____ traversal method should be maintained.

Select the correct option

Preorder

Level order

Postorder

Inorder

Reload Math Equations

Click to Save Answer & Move to Next Question

Learn as 4 Help by Waqas Qureshi 0342176143

Analyze the given code carefully and identify which type of binary tree traversal this is :

```
void traversal (TreeNode<int>* treeNode)
{
    If( treeNode != NULL )
    traversal (treeNode->getLeft());
    cout << "(treeNode->getInfo())<<" ;
    traversal (treeNode->getRight());
}
```

Select the correct option

☐ Preorder

☐ Postorder

☐ Levelorder

☒ Inorder

Click to Save Answer & Move to Next Question

Question # 4 of 10 (Start time: 12:08:45 AM, 23 May 2024) Total Marks: 1

$\text{if} (\text{this} \rightarrow \text{left} == \text{NULL} \ \&\& \ \text{this} \rightarrow \text{right} == \text{NULL})$

If the above condition is satisfied then the current node is:

Select the correct option

Reload Math Equations

☐ Left node

☒ Leaf node

☐ Root node

☐ Right node

Click to Save Answer & Move to Next Question

Question # 5 of 10 (Start time: 12:09:48 AM, 23 May 2024) Total Marks: 1

If p is pointing to a node in an existing tree then $\text{info}(p)$ means:

Select the correct option

Reload Math Equations

☐ Returns true if it is a parent node

☐ Returns false if it is not a parent node

☒ Returns the contents of node p

☐ Returns the pointer of node p

Click to Save Answer & Move to Next Question

If there is a strictly complete binary tree of depth 3, the total number of nodes in it will be?

Select the correct option

- ☐ 15
- ☐ 7
- ☐ 8
- ☒ 15

Question # 7 of 10 (Start time: 12:11:35 AM, 23 May 2024)

Total Marks: 1

Analyze the given code carefully and choose the statement which will be used if we want to set 7 as the value of root ?

```
int x [] = {14,15,4,9,7,18,3,5,16,4,20,17};
```

```
TreeNode * root = new TreeNode();
```

Select the correct option

Reload Math Equations

- ☐ root->setInfo(&x[0]);
- ☒ root->setInfo(&x[4]);
- ☐ root->setInfo(&x);
- ☐ root->setInfo(x);

Question # 8 of 10 (Start time: 12:12:19 AM, 23 May 2024) Total Marks: 1

Which one of the following case is the most complicated case to delete a node from BST?

Select the correct option

☐ No case is complicated.

☒ Node to be deleted has either left child or right child

☐ Node to be deleted is the leaf node

☐ Node to be deleted has both the left and right children

Click to Save Answer & Move to Next Question

Question # 9 of 10 (Start time: 12:13:09 AM, 23 May 2024) Total Marks: 1

What will be output of the given C++ code?

```
char lhs[] = "Army";  
char rhs[] = "Army";  
cout<<strcmp(lhs,rhs);
```

Select the correct option

☐ Negative number

☐ Syntax error

☐ Positive number

☒ 0

CS301 - Data Structures (Graded Quiz No. 2) Quiz Start Time: 12:05 AM, 23 May 2024

Question # 10 of 10 (Start time: 12:14:14 AM, 23 May 2024) Total Marks: 1

_____ is an area in computer memory that is used to allocate memory dynamically.

Select the correct option

☐ Static

☒ Heap

☐ Global

☐ Stack

Reload Math Equations

The _____ sign before the name of the variable means that the address of the variable is being passed :

Select the correct option

☐ ||

☐ =

☐ #

☒ &

Which operation of queue data structure is used to insert an element into the Queue?

Select the correct option

- ☐ front()
- ☒ enqueue()
- ☐ remove()
- ☐ dequeue()

Question # 4 of 10 (Start time: 03:25:31 PM, 23 May 2024) Total Marks: 1

Which of the following is a non-linear data structure?

Select the correct option

- ☒ Tree
- ☐ Queue
- ☐ Linked List
- ☐ Stack

Question # 5 of 10 (Start time: 03:25:52 PM, 23 May 2024) Total Marks: 1

What will be level order traversal of the given BST ?

```
graph TD; 6((6)) --- 2((2)); 6 --- 8((8)); 2 --- 1((1)); 2 --- 5((5)); 5 --- 3((3)); 5 --- 4((4));
```

Select the correct option

☒ 6 2 8 1 5 3 4

☐ 1 4 3 5 2 8 6

☐ 1 2 3 4 5 6 8

Question # 6 of 10 (Start time: 03:27:36 PM, 23 May 2024) Total Marks: 1

During in-order traversal using recursive calls, if we found a node is NULL. It means this node will satisfy following condition.

Select the correct option

☒ It will not have both left and right children

☐ It will have only left child

☐ It will not have right child

☐ It will not have left child

Click to Save Answer & Move to Next Question

Question # 7 of 10 (Start time: 03:29:29 PM, 23 May 2024) Total Marks: 1

In which traversal method, the recursive calls can be used to traverse a binary tree ?

Select the correct option

☐ In postorder and preorder traversal only

☐ In preorder and inorder traversal only

☒ In preorder, postorder and inorder traversal only

☐ In postorder and inorder traversal only

Click to Save Answer & Move to Next Question

Question # 8 of 10 (Start time: 03:30:57 PM, 23 May 2024) Total Marks: 1

For making Binary Search Tree for Strings we need, _____ data type.

Select the correct option

☒ char

☐ double

☐ float

☐ int

CS301 - Data Structures (Graded Quiz No. 2) Quiz Start Time: 03:24 PM, 23 May 2024

Question # 9 of 10 (Start time: 03:32:25 PM, 23 May 2024) Total Marks: 1

_____ is an area in computer memory that is used to allocate memory dynamically.

Select the correct option

☒ Heap

☐ Global

☐ Stack

☐ Static

Reload Math Equations

A queue is a data structure where elements are

Select the correct option

☐ inserted and removed from both ends.

☒ inserted at the back and removed from the front.

☐ inserted and removed from the top.

☐ inserted at the front and removed from the back.